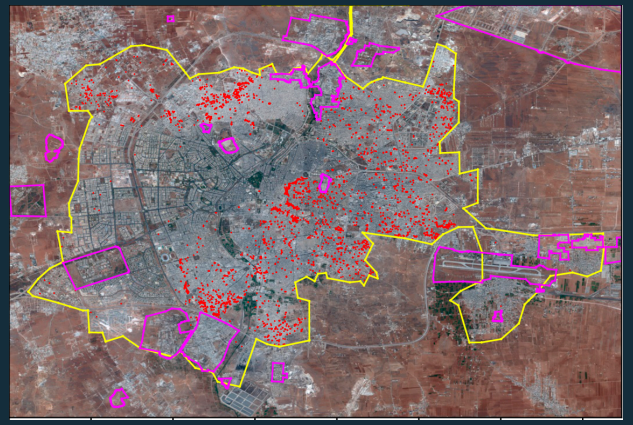
Monitoring war destruction from space using machine learning

Mueller et al. · PNAS, 2021

Case study: Building destruction during the Syrian civil war



Aleppo · Fig. 1A, Mueller et al. (2021)
Imagery: Google Earth/Maxar; labels: UNITAR/UNOSAT

Aurel Kelterborn, Basil Kohler
Paper presentation · 30 October 2026

Why measure destruction from space?

Destruction data are scarce

Existing data: slow, costly, incomplete
Needed for aid, human rights and research

Destruction is rare

Aleppo: only 2.8% of patches show destruction
Small error rates create many false alarms

14,850

false alarms vs. 900 hits
100,000 patches, 1% destroyed,
90% recall, 15% FPR

Research question

Can repeated satellite images and machine learning reliably map where and when buildings are destroyed?

Source: Introduction; "Why Automated War-Destruction Monitoring Is Hard"

Data: repeated images, sparse annotations

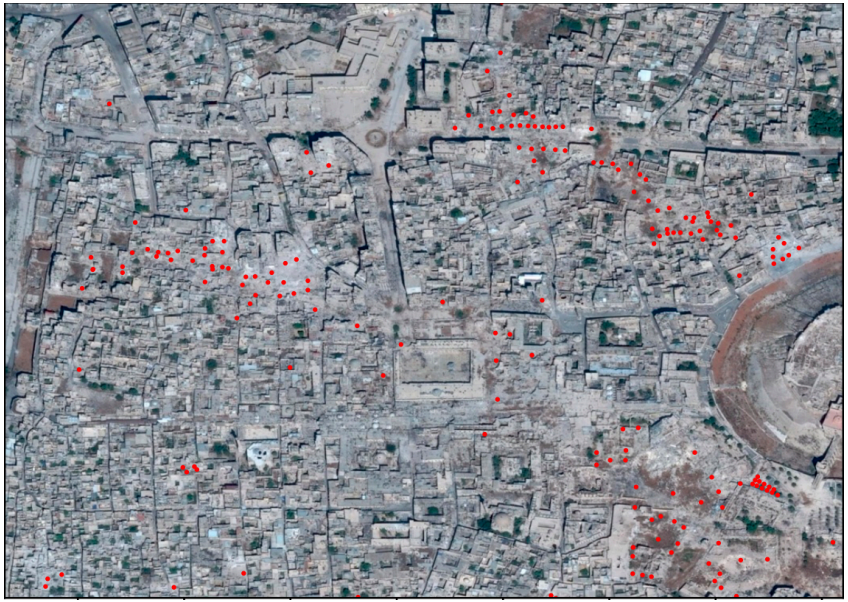
6 Syrian cities · 2011–2017

RGB imagery at approximately 50 cm/pixel

64 × 64 pixel patches ≈ 32 × 32 m

UNOSAT labels: completely destroyed

Aleppo: 22 later images but only 4 annotation dates



Red points: UNOSAT destruction annotations

Source: Methods and Table 1; image: Fig. 1B

More training labels through persistence

Assumption: destruction persists during the study period

Earlier annotation	Images between	Later annotation
Not destroyed	Not destroyed	Not destroyed
Destroyed	Destroyed	Destroyed
Not destroyed	Unknown	Destroyed

About 2.3 million labeled patch–date observations

Unknown entries receive model predictions later

Source: Contextual Label Augmentation and Fig. 2; table is an illustrative example

Two stages: visual evidence and context

01 **CNN**

Reference + later image patch

Initial destruction score

02 **Random forest**

CNN scores across space and time

Refined destruction score

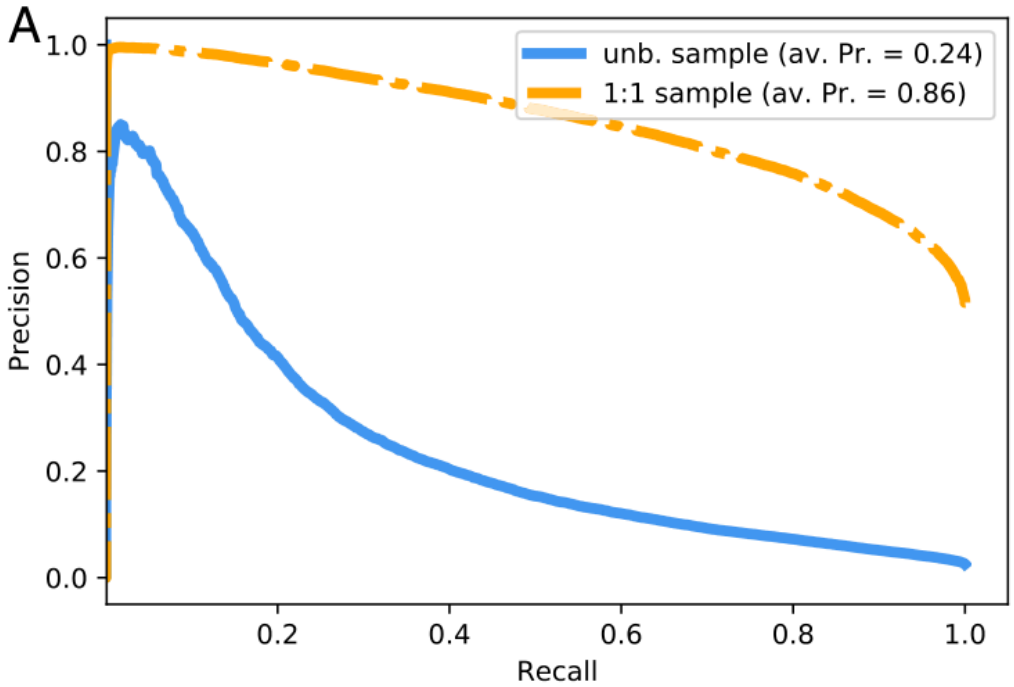
Temporal context includes 2 earlier and 2 later observations

Links to class

CNN learns visual features itself, no hand-crafted rules for rubble (Session 1)
70/30 split by location, like the mugshot paper splits by arrestee (Session 2)
Two stages stack a CNN with trees, like the mugshot predictor (Session 2)

Source: CNN Architecture and Two-Stage Classification Procedure

Results: Performance under class imbalance



Precision

Correct alerts / all alerts

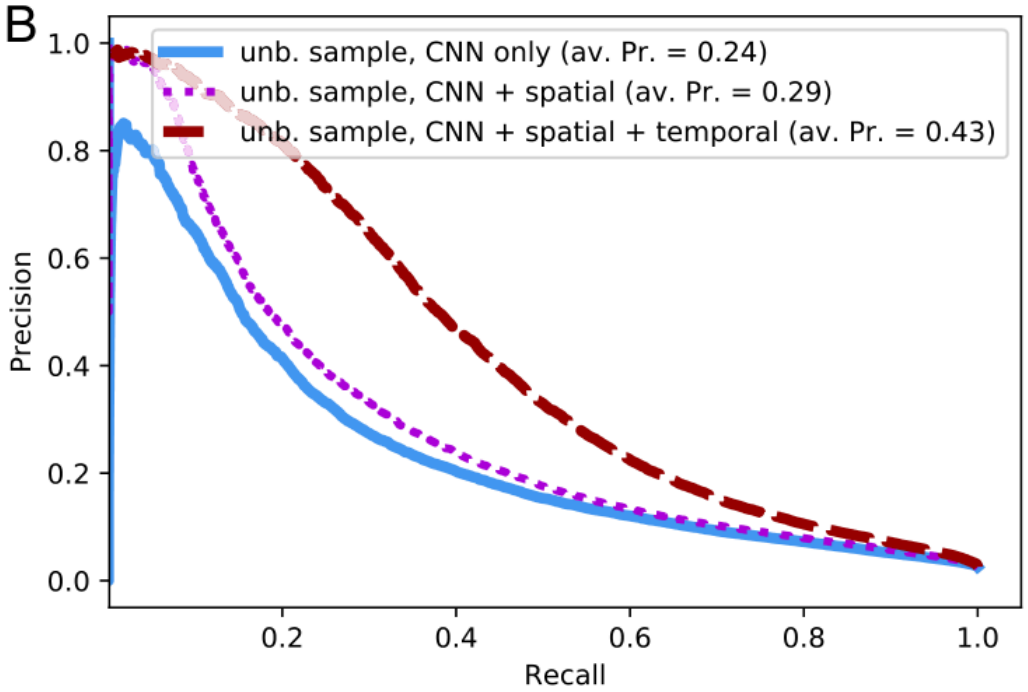
Recall

Detected destruction / all actual destruction

Average precision 86% balanced / 24% actual

Source: Results, Overall Performance; original exhibit: Fig. 3A

Temporal context improves detection



Average precision

24.5%

CNN alone

28.7%

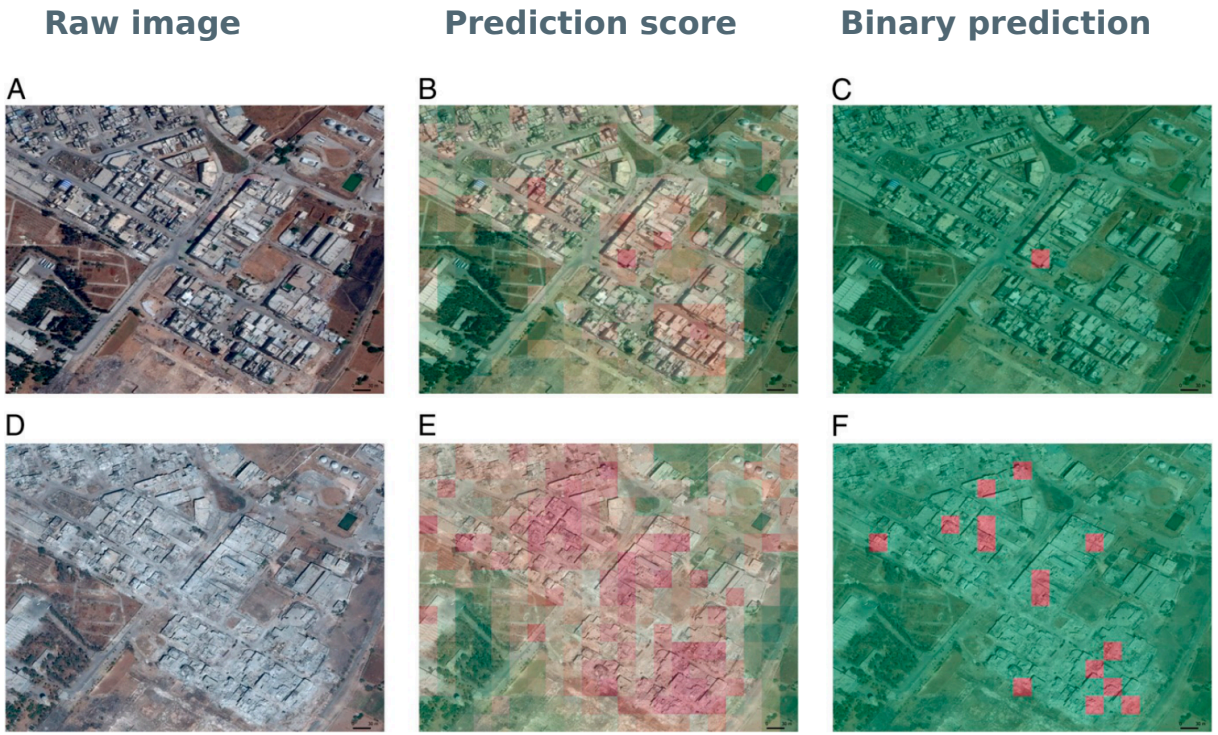
+ spatial context

42.5%

+ spatial and temporal context

Source: Fig. 3B and Table 2; AP = average precision across thresholds

Validation beyond the training locations



6 Dec 2016: before
18 Dec 2016: after

Independent timing check

731 bombing reports

Scores increase after reported bombings

Source: External Validation Exercises; original exhibit: Fig. 4, Ramouse, Aleppo

How convincing are the results?

Strong retrospective evidence

Realistic class proportions
Tested on unseen area + bombings

Real-time use untested

Best model uses 2 future images
Precision 0.43: humans must verify

Consistent improvement

context improves performance
precision rises from 24.5% to 42.5%

Low precision in some cities

With 1% of buildings destroyed,
average precision remains 11.7%

Research, policy and ethical implications

Research

Study destruction patterns and links with displacement or conflict dynamics

Humanitarian use

Prioritize human verification and support reconstruction assessments

Ethics and future work

Sensitive maps can expose vulnerable areas
Compare immediate alerts with later, more reliable revisions

Source: Introduction and Discussion

Discussion

1 Should detailed destruction maps be publicly available if they could help humanitarian teams but also be misused by armed groups?

2 Should the classifier prioritize high recall to detect more destroyed buildings, or high precision to minimize false alarms?